

1. Algorithm

Main Program Algorithm

1. Start
2. Display menu options to the user (1 to 5 for patterns, 6 to Exit).
3. Read user's choice into choice.
4. If choice == 6, exit the program.
5. If choice is between 1 and 5:
 - Prompt user to enter the number of rows n.
 - Read n.
6. Use switch(choice):
 - Case 1: Call full_pyramid()
 - Case 2: Call inverted_pyramid()
 - Case 3: Call diamond_pattern()
 - Case 4: Call palindromic_pyramid()
 - Case 5: Call repeated_number_pyramid()
 - Default: Print "Invalid Choice".
7. Stop

Function Algorithms

A. full_pyramid()

1. Loop i from 1 to n:
 - Loop j from 1 to (n - i): Print space " ".
 - Loop k from 1 to (2*i - 1): Print "*".
 - Print new line.

B. inverted_pyramid()

1. Loop i from n down to 1:
 - Loop j from 1 to (n - i): Print space " ".
 - Loop k from 1 to (2*i - 1): Print "*".
 - Print new line.

C. diamond_pattern()

1. Upper Half: Loop i from 1 to n:
 - Loop j from 1 to (n - i): Print space " ".
 - Loop k from 1 to (2*i - 1): Print "*".
 - Print new line.
2. Lower Half: Loop i from n - 1 down to 1:
 - Loop j from 1 to (n - i): Print space " ".
 - Loop k from 1 to (2*i - 1): Print "*".
 - Print new line.

D. palindromic_pyramid()

1. Loop i from 1 to n:
 - Loop j from 1 to (n - i): Print double space " ".

- Loop k from 1 to i: Print k with space.
- Loop k from i - 1 down to 1: Print k with space.
- Print new line.

E. repeated_number_pyramid()

1. Loop i from 1 to n:
 - Loop j from 1 to (n - i): Print double space " ".
 - Loop k from 1 to (2*i - 1): Print i with space.
 - Print new line.

2. Test Cases

Test Case ID	Input (Choice, n)	Expected Output Summary	Status
TC_01	Choice = 1, n = 4	Full pyramid of 4 rows made of '*'	PASS
TC_02	Choice = 2, n = 4	Inverted pyramid of 4 rows made of '*'	PASS
TC_03	Choice = 3, n = 4	Diamond pattern (Upper half 4 rows, Total 7 rows)	PASS
TC_04	Choice = 4, n = 4	Palindromic number pyramid with 4 rows (1 to 4 and back)	PASS
TC_05	Choice = 5, n = 4	Repeated number pyramid (1, 2, 3, 4 printed (2n-1) times)	PASS
TC_06	Choice = 7	Displays "Invalid Choice" message	PASS

3. C Source Code

```
#include <stdio.h>

void full_pyramid();
void inverted_pyramid();
void diamond_pattern();
void palindromic_pyramid();
void repeated_number_pyramid();

int n;

int main() {
    int choice;

    printf("===== MENU =====\n");
    printf("1. Full Pyramid\n");
    printf("2. Inverted Pyramid\n");
    printf("3. Diamond Pattern\n");
    printf("4. Palindromic Number Pyramid\n");
    printf("5. Repeated-Number Pyramid\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    if (choice >= 1 && choice <= 5) {
        printf("Enter the number of rows (n): ");
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        scanf("%d", &n);
    }

    switch (choice) {
        case 1:
            full_pyramid();
            break;
        case 2:
            inverted_pyramid();
            break;
        case 3:
            diamond_pattern();
            break;
        case 4:
            palindromic_pyramid();
            break;
        case 5:
            repeated_number_pyramid();
            break;
        default:
            printf("Invalid choice! Please select between 1 and 5.
");
    }

    return 0;
}

void full_pyramid() {
    int i, j, k;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= 2 * i - 1; k++) {
            printf("*");
        }
        printf("
");
    }
}

void inverted_pyramid() {
    int i, j, k;
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= 2 * i - 1; k++) {
            printf("*");
        }
        printf("
");
    }
}

void diamond_pattern() {
    int i, j, k;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }

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        for (k = 1; k <= 2 * i - 1; k++) {
            printf("*");
        }
        printf("
");
    }
    for (i = n - 1; i >= 1; i--) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= 2 * i - 1; k++) {
            printf("*");
        }
        printf("
");
    }
}

void palindromic_pyramid() {
    int i, j, k;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= i; k++) {
            printf("%d ", k);
        }
        for (k = i - 1; k >= 1; k--) {
            printf("%d ", k);
        }
        printf("
");
    }
}

void repeated_number_pyramid() {
    int i, j, k;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= 2 * i - 1; k++) {
            printf("%d ", i);
        }
        printf("
");
    }
}

```